Simplified Embedding Scheme for Quantum Annealing Applied to Activity Detection in Massive Wireless Networks

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Outline

1. Activity detection in wireless networks
2. Quantum computing for signal processing
3. Hardware-aware implementation

Non-orthogonal multiple access (NOMA)

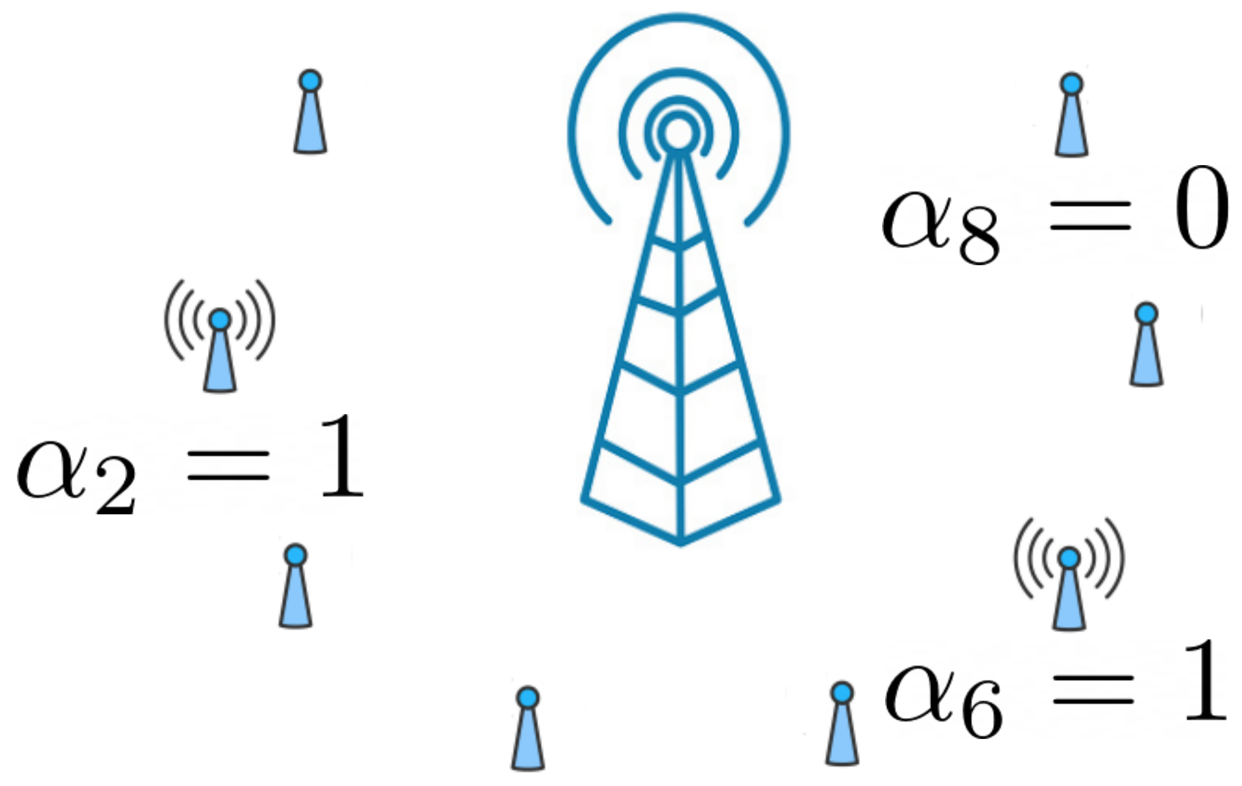
☞ Several user equipments wish to transmit data to a base station : need for an *access protocol*

☞ Sporadic trafic pattern: **activity detection**

☞ **NOMA:** assign *non-orthogonal* preambles amongst the user's equipments

System model

- Each user equipment is assigned a unique spreading code. *The active users send their code*



- Best guess? Go with Maximum Likelihood:

$$\boxed{\hat{\boldsymbol{\alpha}} = \arg\min_{\boldsymbol{\alpha}\in\{0,1\}^N} \mathcal{L}(\boldsymbol{\alpha})}$$

This is a NP-hard computation !

An intuitive approach: quantum annealing

- A physical system always tends to *minimize its energy*.

- Quantum annealing solves optimization problems via energy minimization

$$H(t) = \mathcal{A}(t)H_C + \mathcal{B}(t)H_P$$

Driving term
Trivial ground state

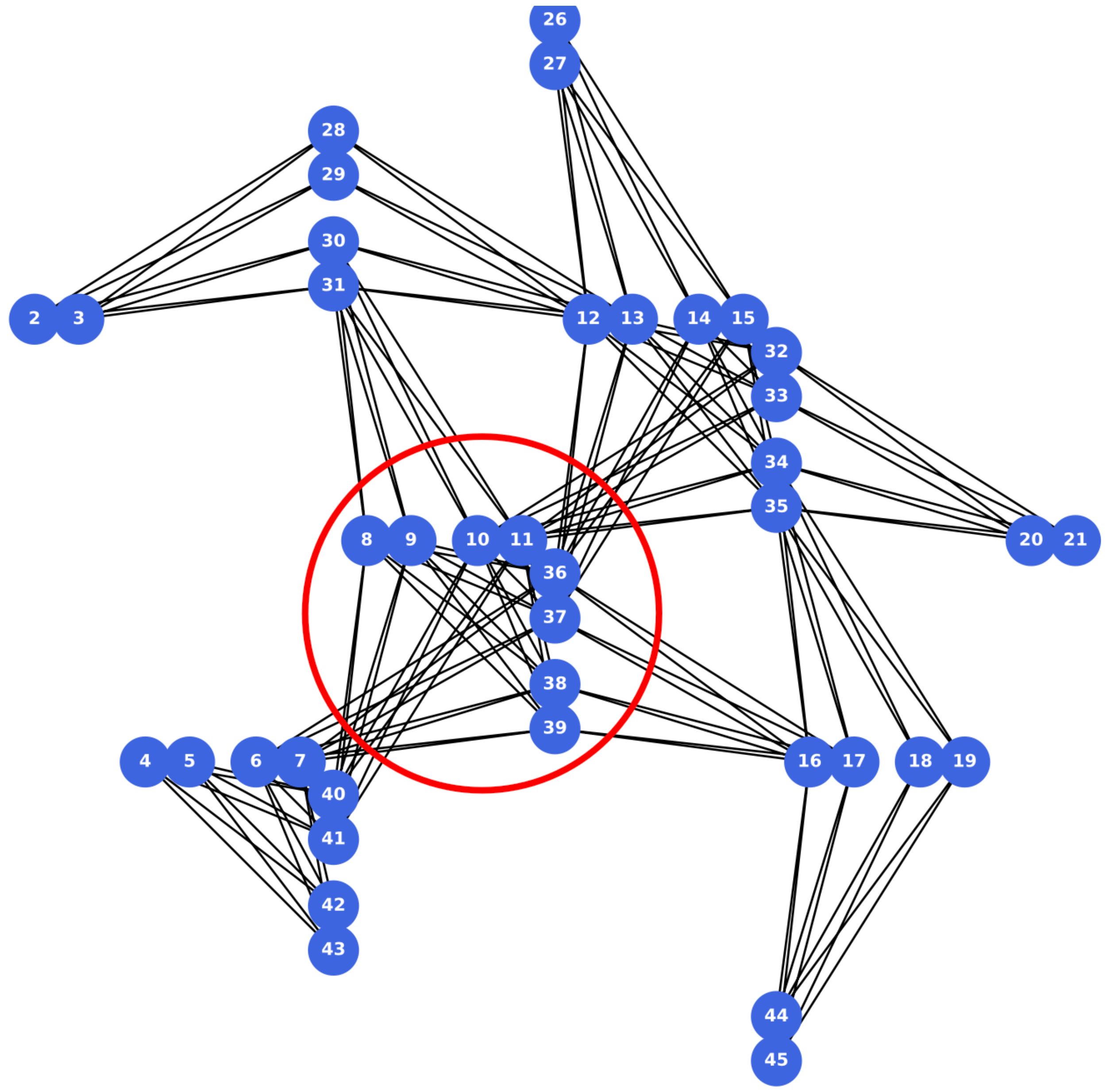
Problem Hamiltonian
Solution of the problem

H_C	Easy low-energy configuration, induce quantum fluctuations between configurations
$\mathcal{A}(t), \mathcal{B}(t)$	Adiabatic ramps, keep the system near the ground state.
H_P	Encodes the problem, its ground state is the state of interest

Prototypal problem Hamiltonian

☞ D-Wave's QPU encodes **Ising Hamiltonian**

$$H_P = -\sum_{i<j} J_{ij} Z_i Z_j - \sum_i b_i Z_i.$$

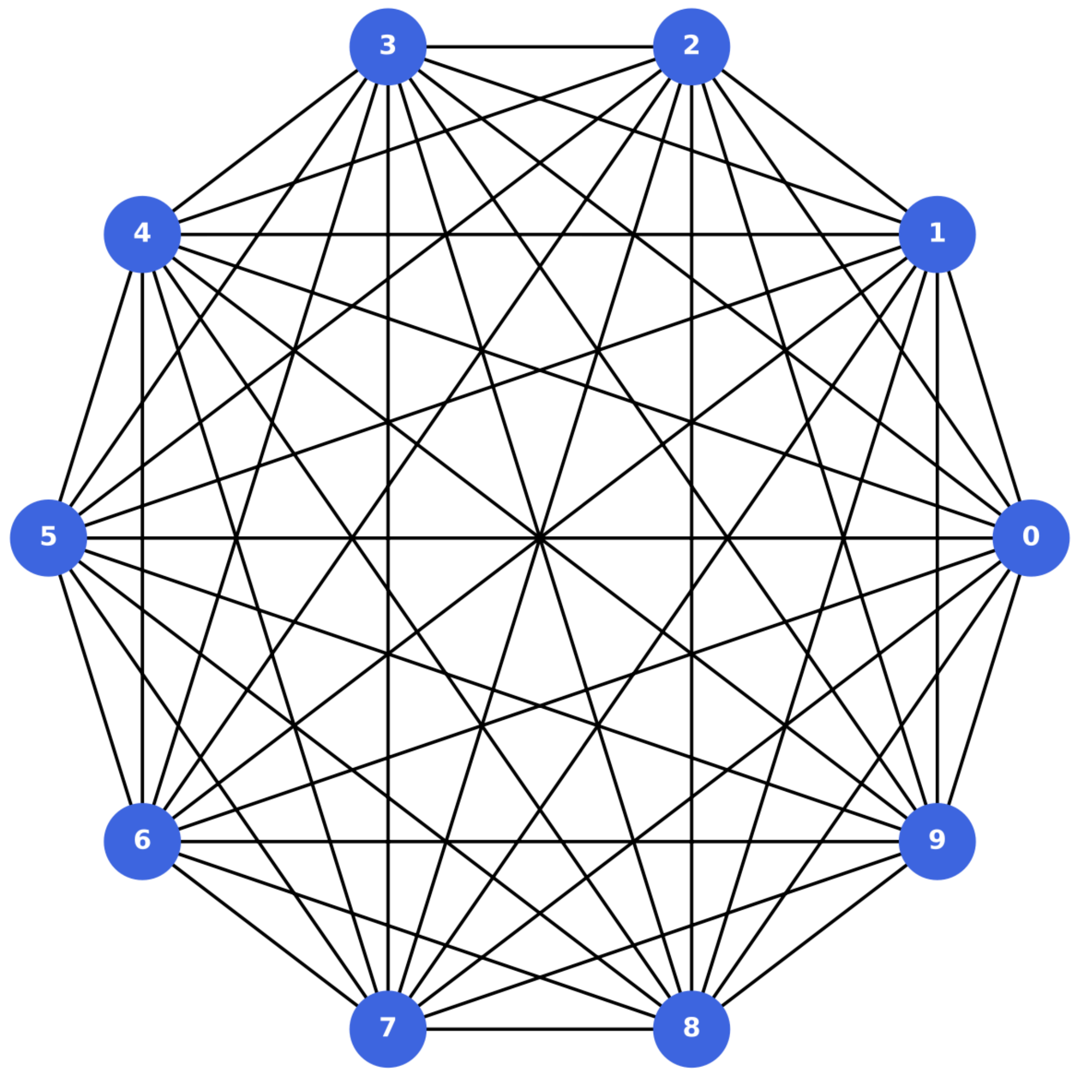


Ising Hamiltonian

☞ The negative log-likelihood is represented by an Ising Hamiltonian

$$H_P = -\sum_{i<j} J_{ij} Z_i Z_j - \sum_i b_i Z_i.$$

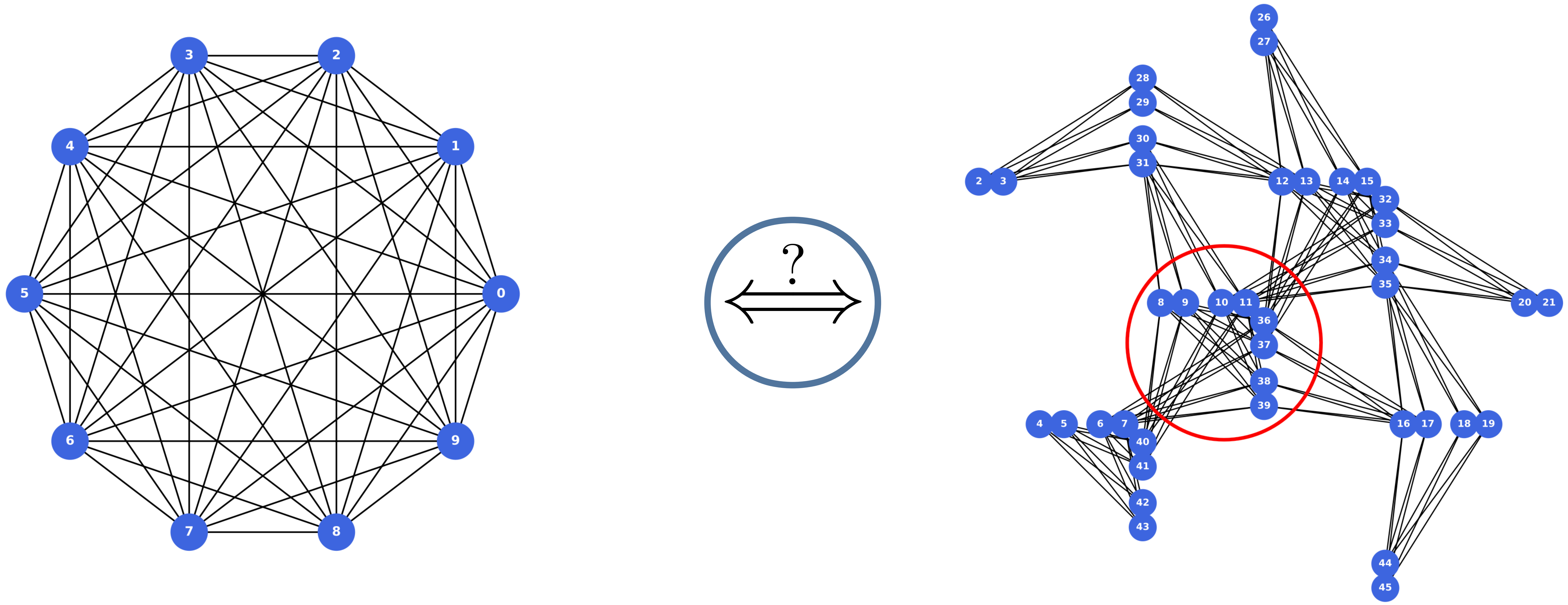
This energy function involves *couplings* and *local fields*, related to the system model



Graph representation

Embedding on the QPU

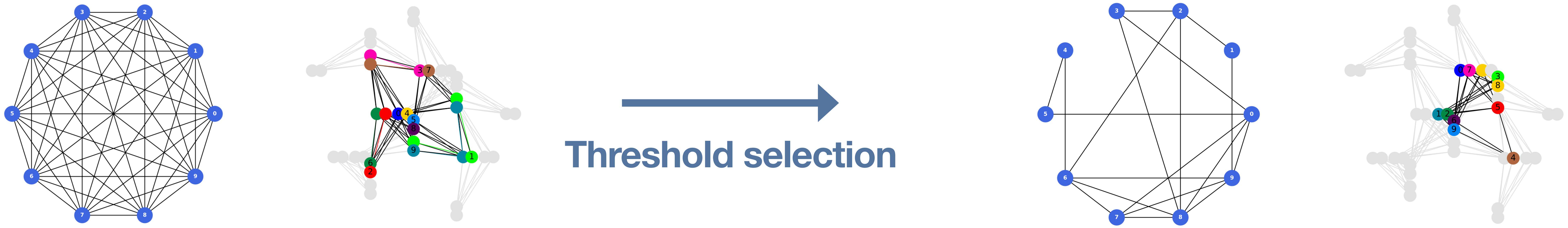
☞ D-Wave's Advantage System runs quantum annealing on quantum hardware



☞ Hardware constraints demand **problem embedding**. But **overhead** is part of the deal.

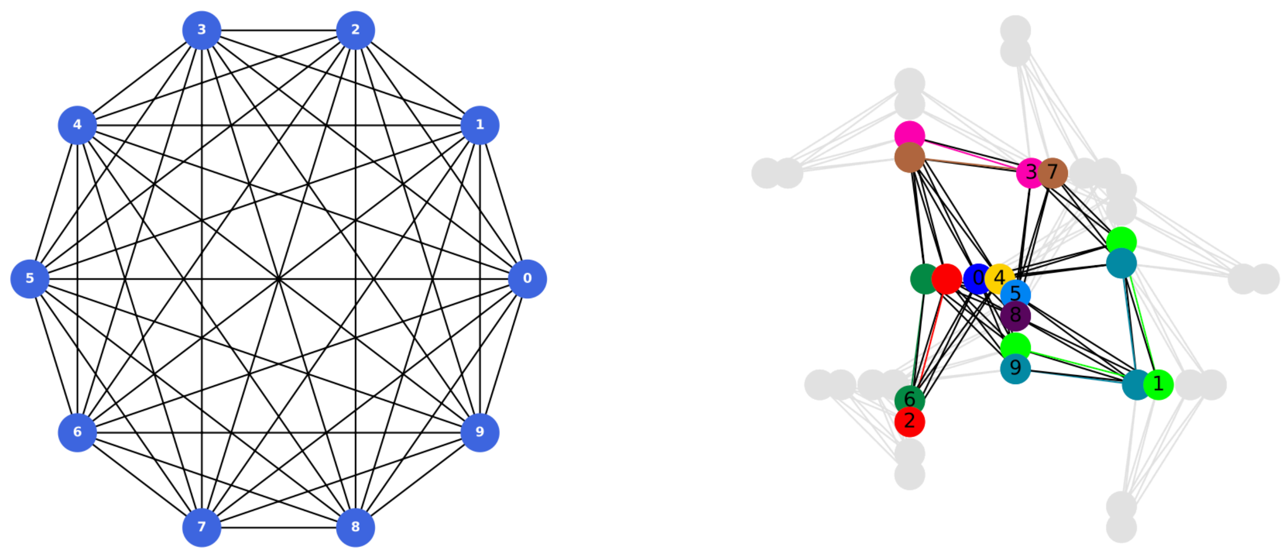
Proposition: the threshold method

☞ Neglect the couplings of lowest magnitude

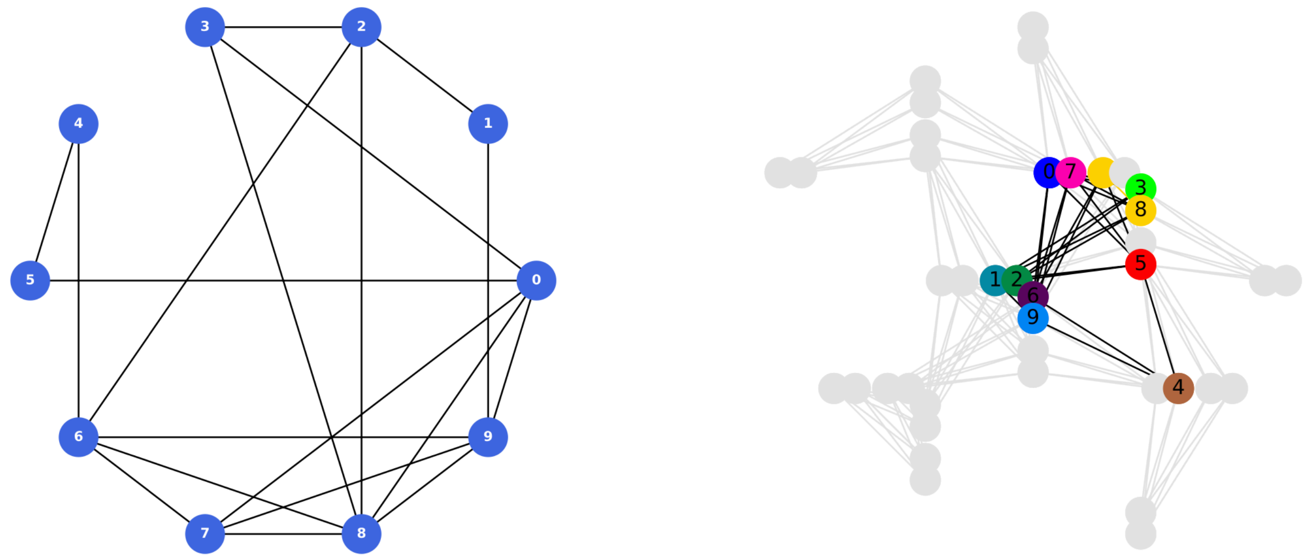


Proposition: the threshold method

☞ Neglect the couplings of lowest magnitude



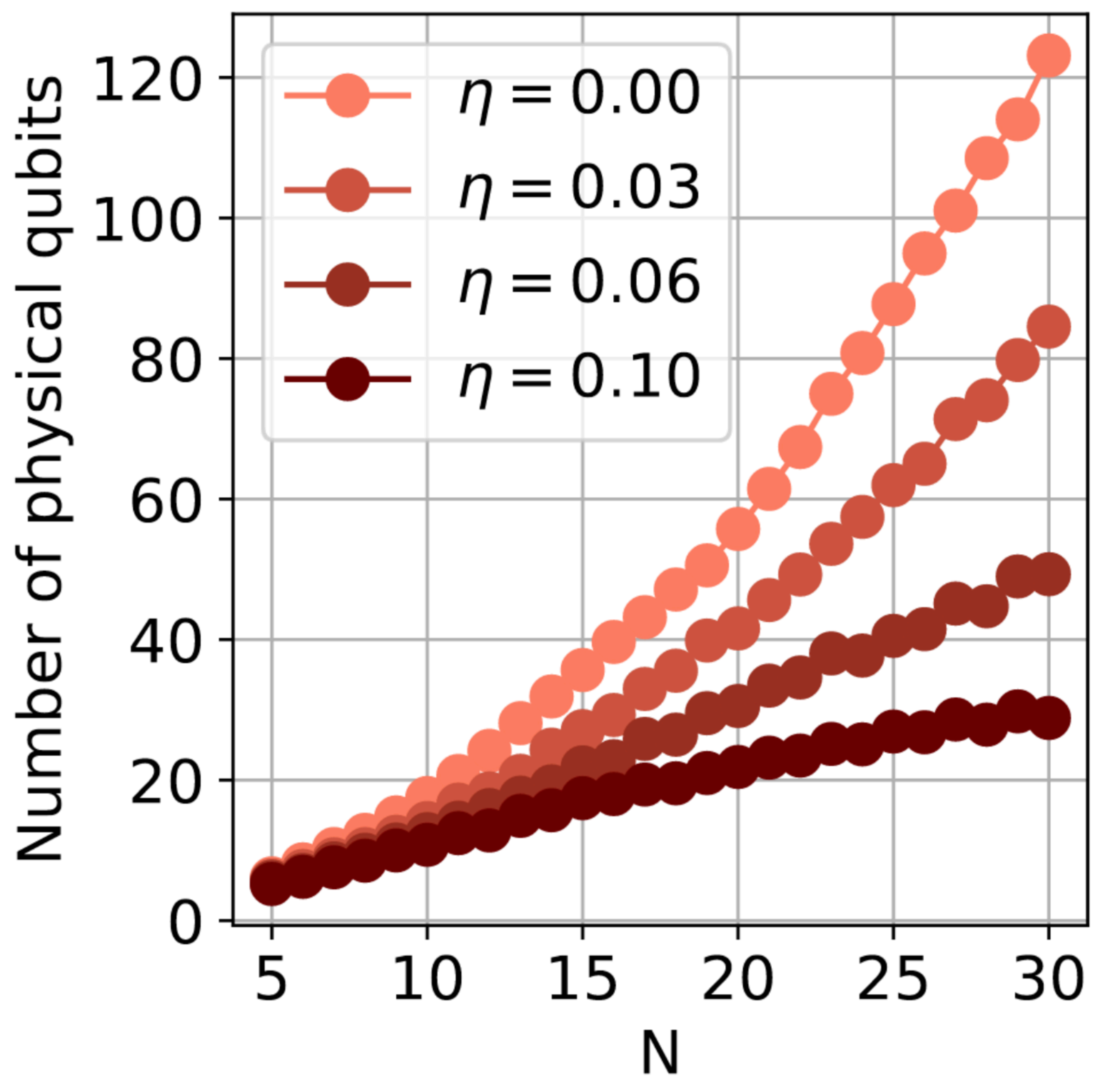
Threshold selection



☞ Trade-off: ressources vs. precision

N Number of sensors

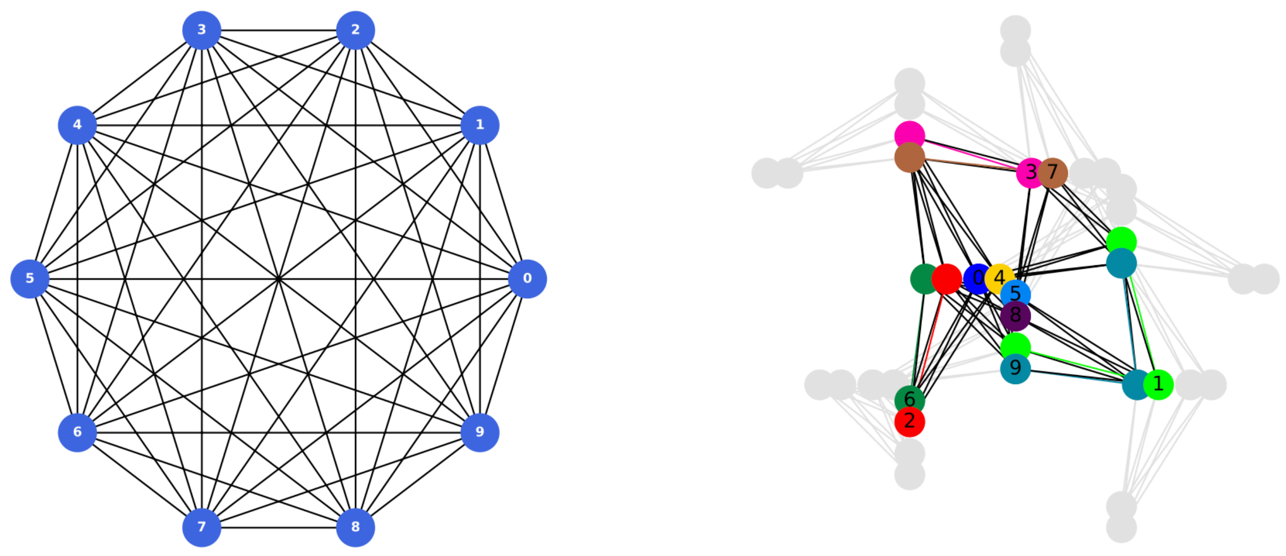
η Threshold value



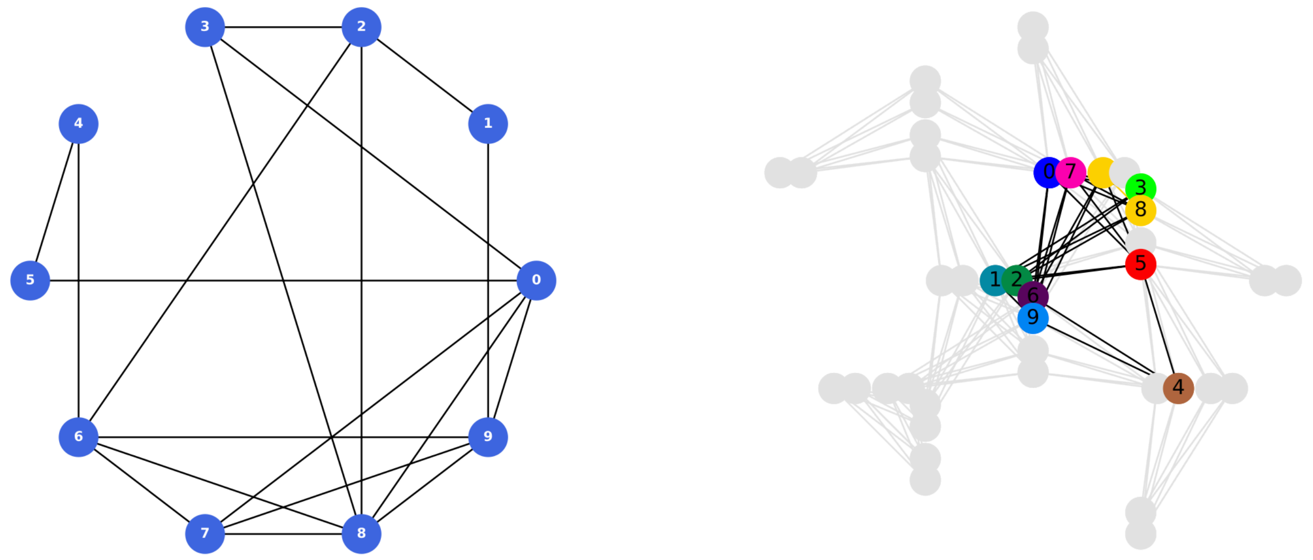
Physical qubit reduction !

Proposition: the threshold method

- Neglect the couplings of lowest magnitude



Threshold selection



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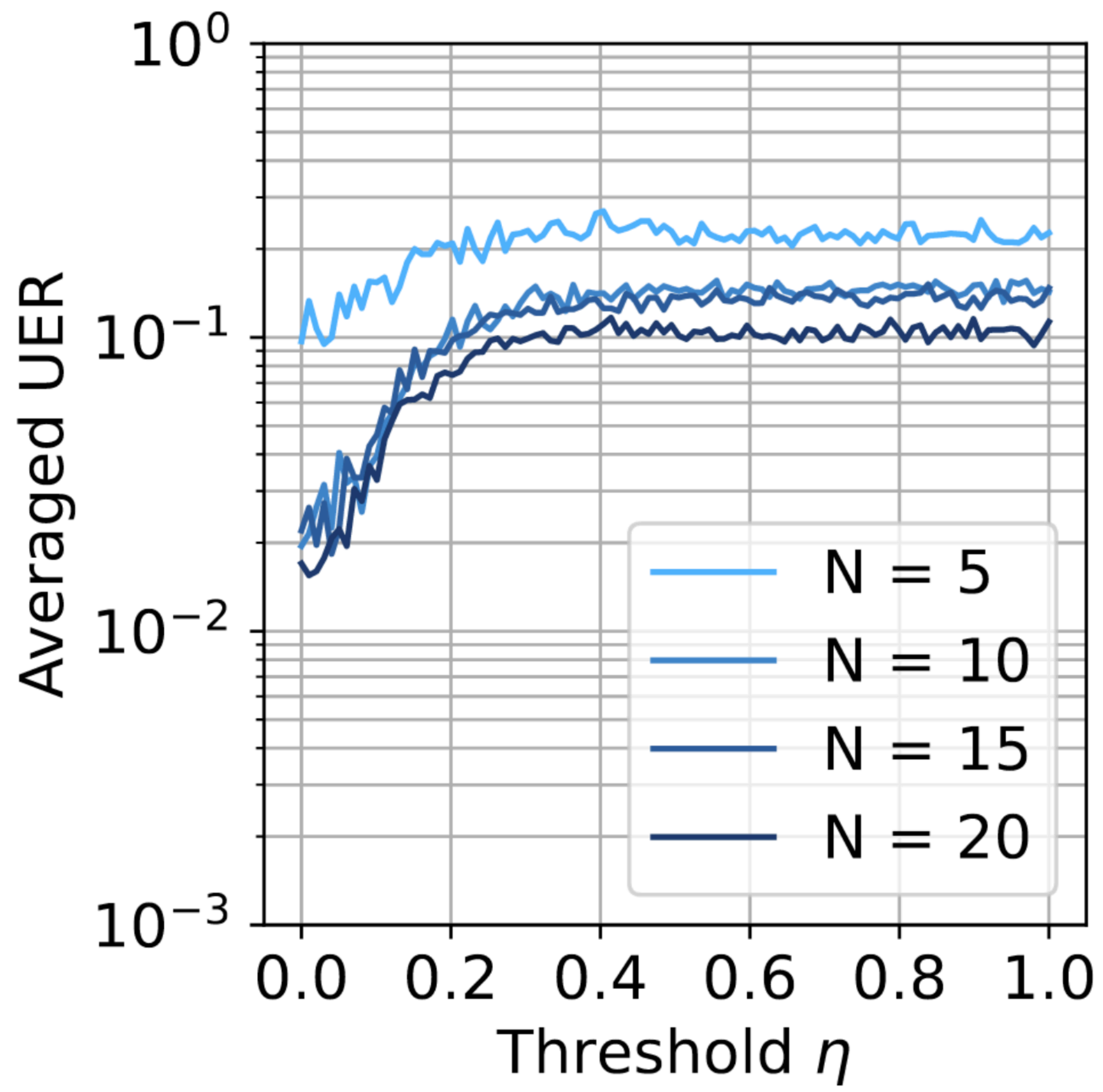


N Number of sensors

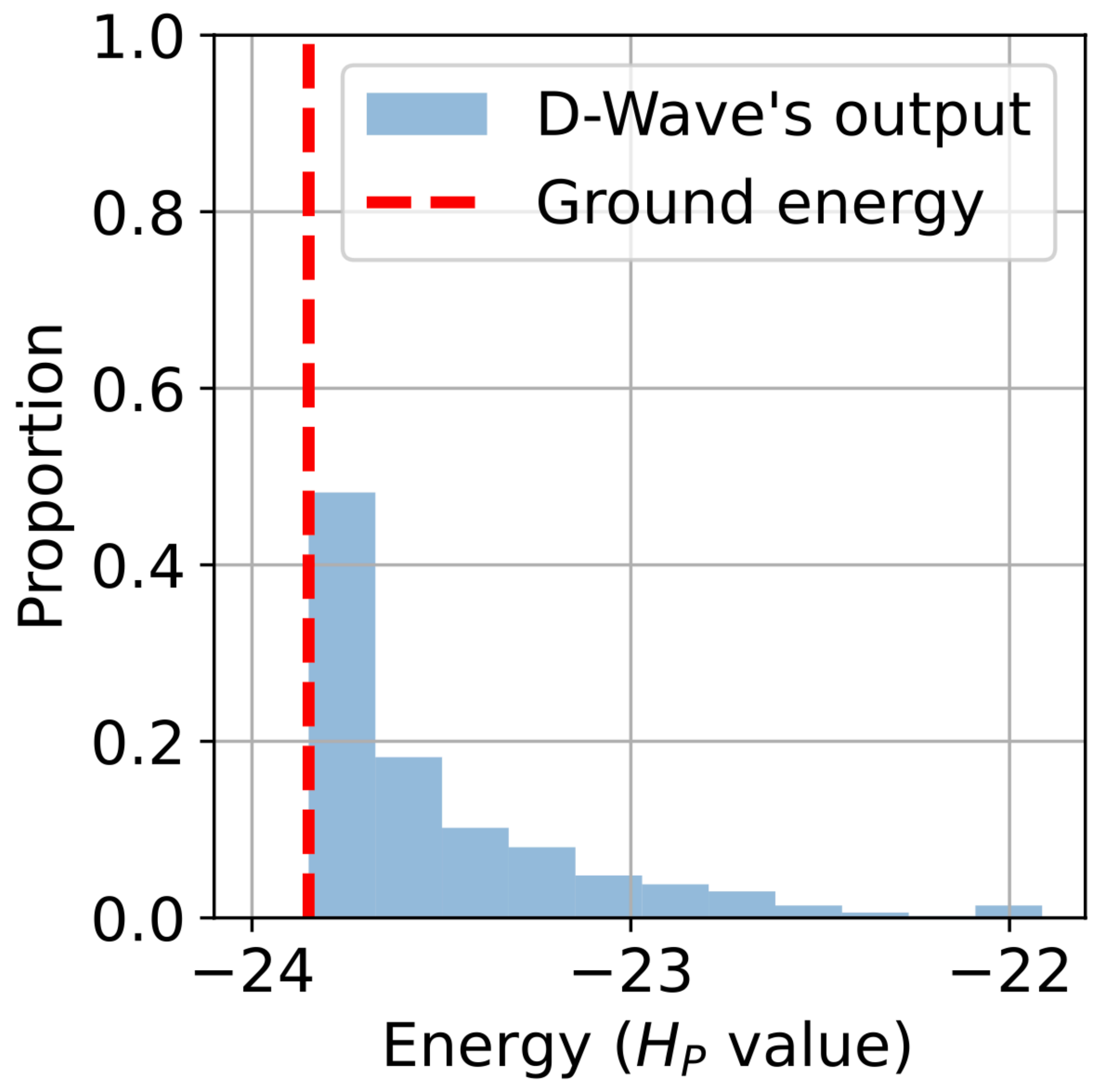
η Threshold value

UER Reliability of the recovery

But … reliability degradation

Method and results

- Experiments conducted on D-Wave's Advantage system6.4
  Annealing time fixed to T = 1 μs



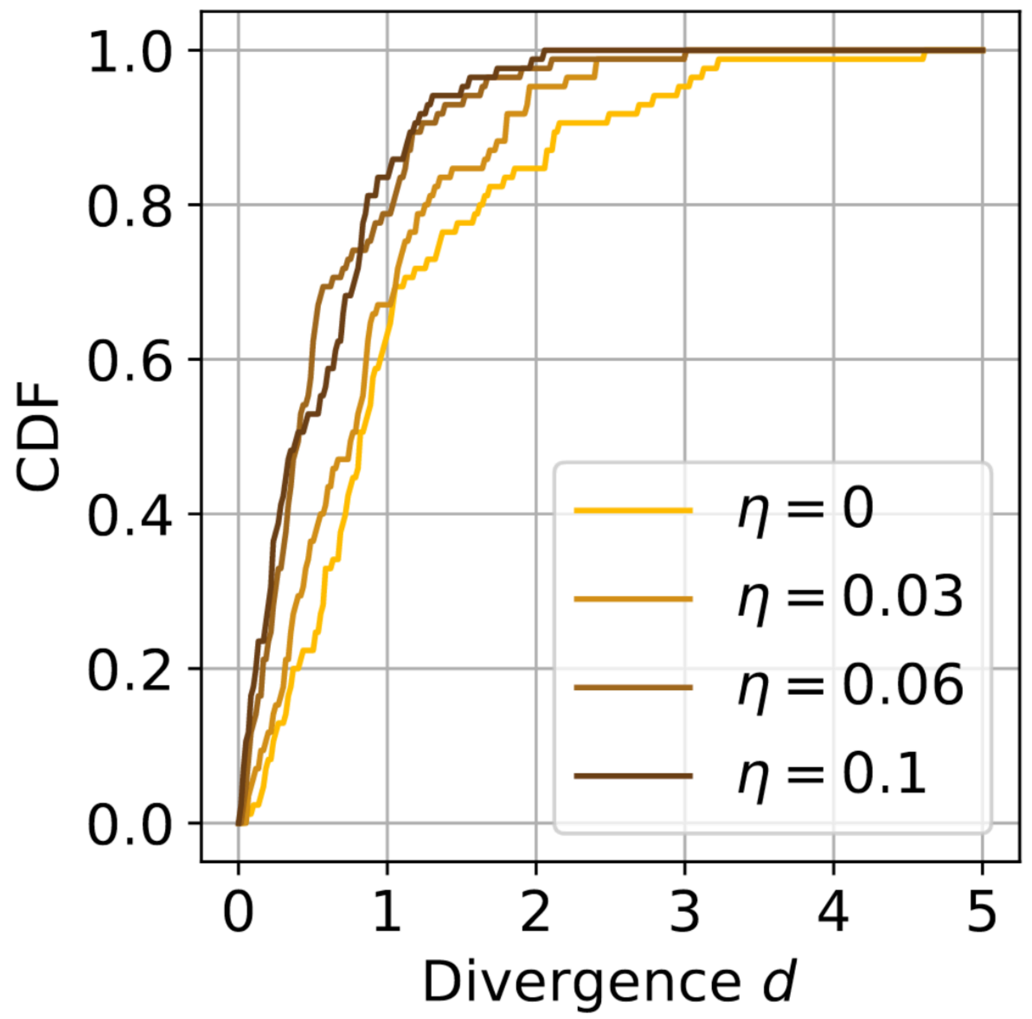
- Run several shots of annealing to *sample* the obtained quantum superposition (at least, approximately…)

Method and results

☞ Convergence of the problems

d Deviation from the ground state

η Threshold value



Cut problems: better convergence !

Method and results

Convergence of the problems

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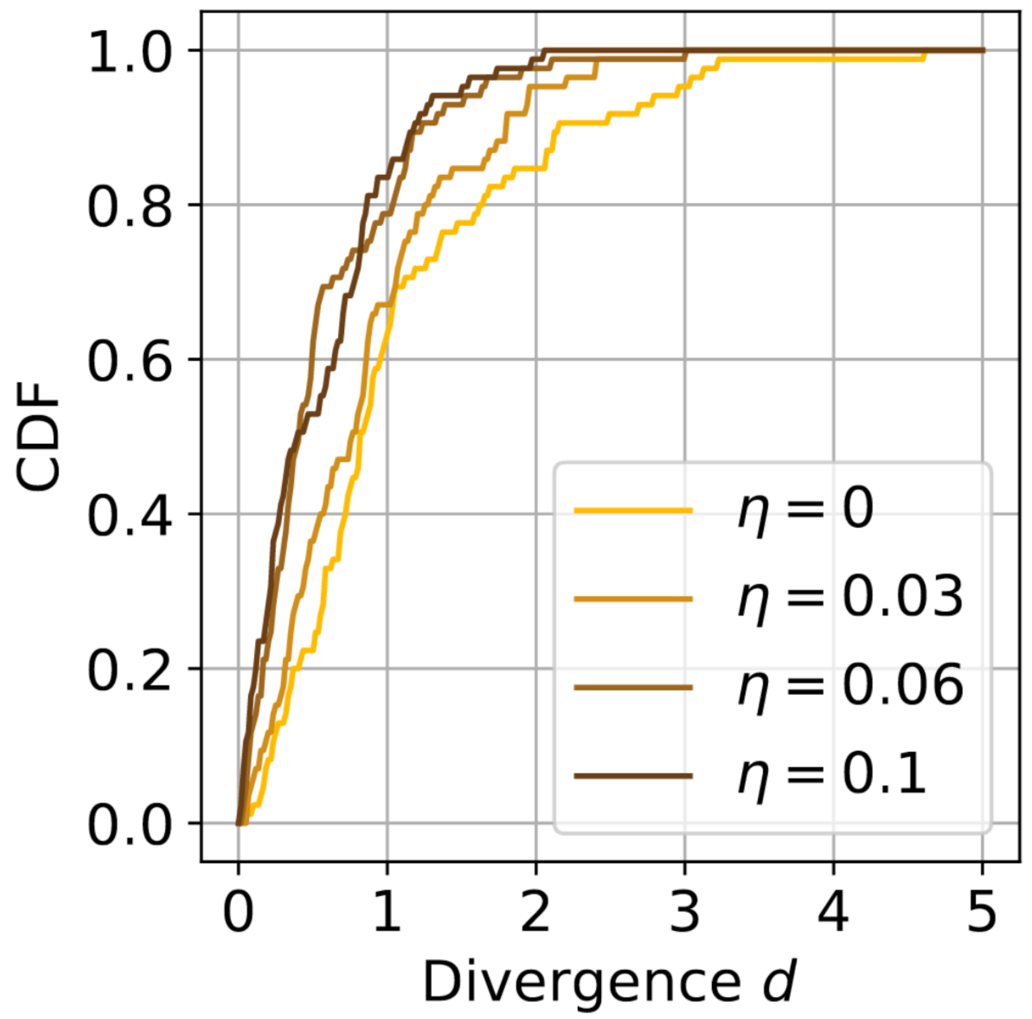
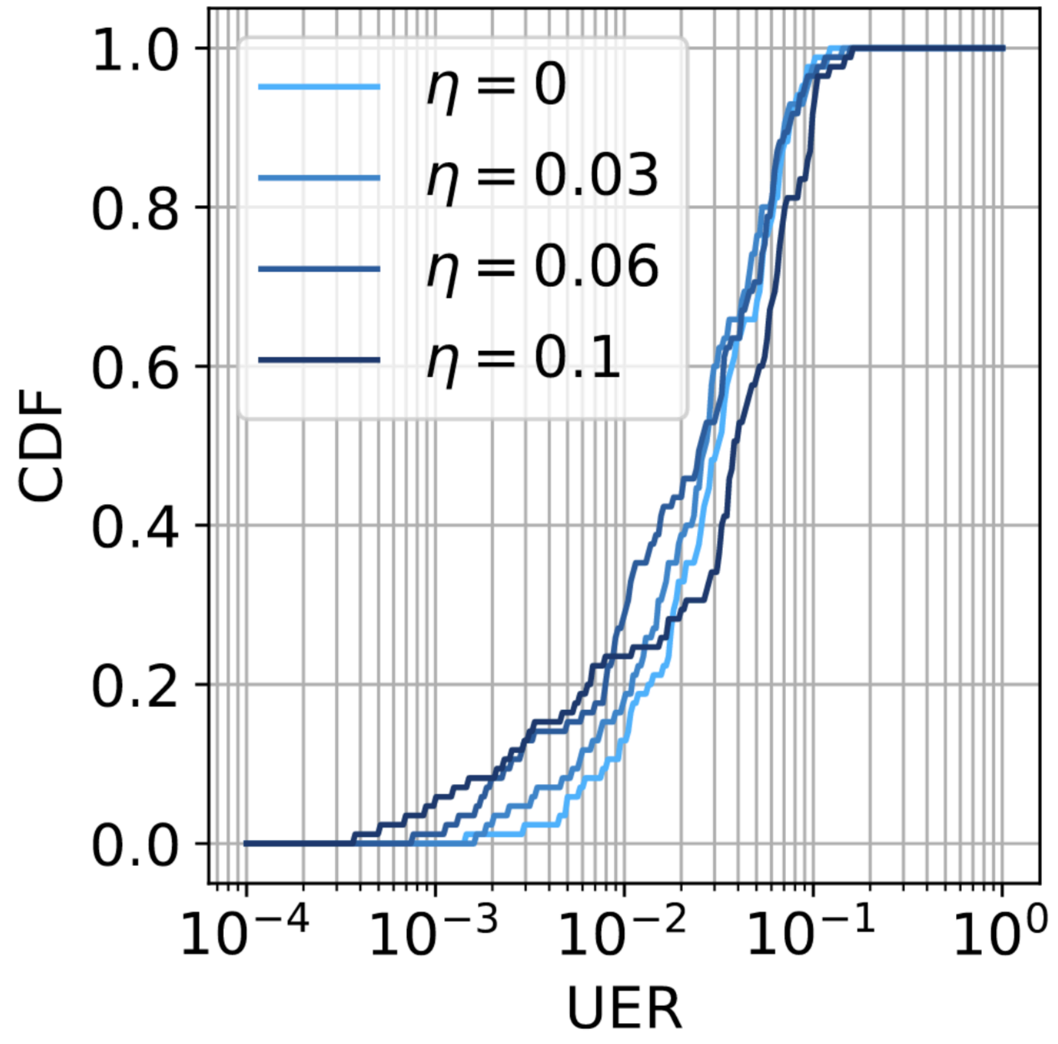


Cut problems: better convergence !

Reliability of the solutions

UER Reliability of the recovery

η Threshold value



Enhanced reliability at key threshold values

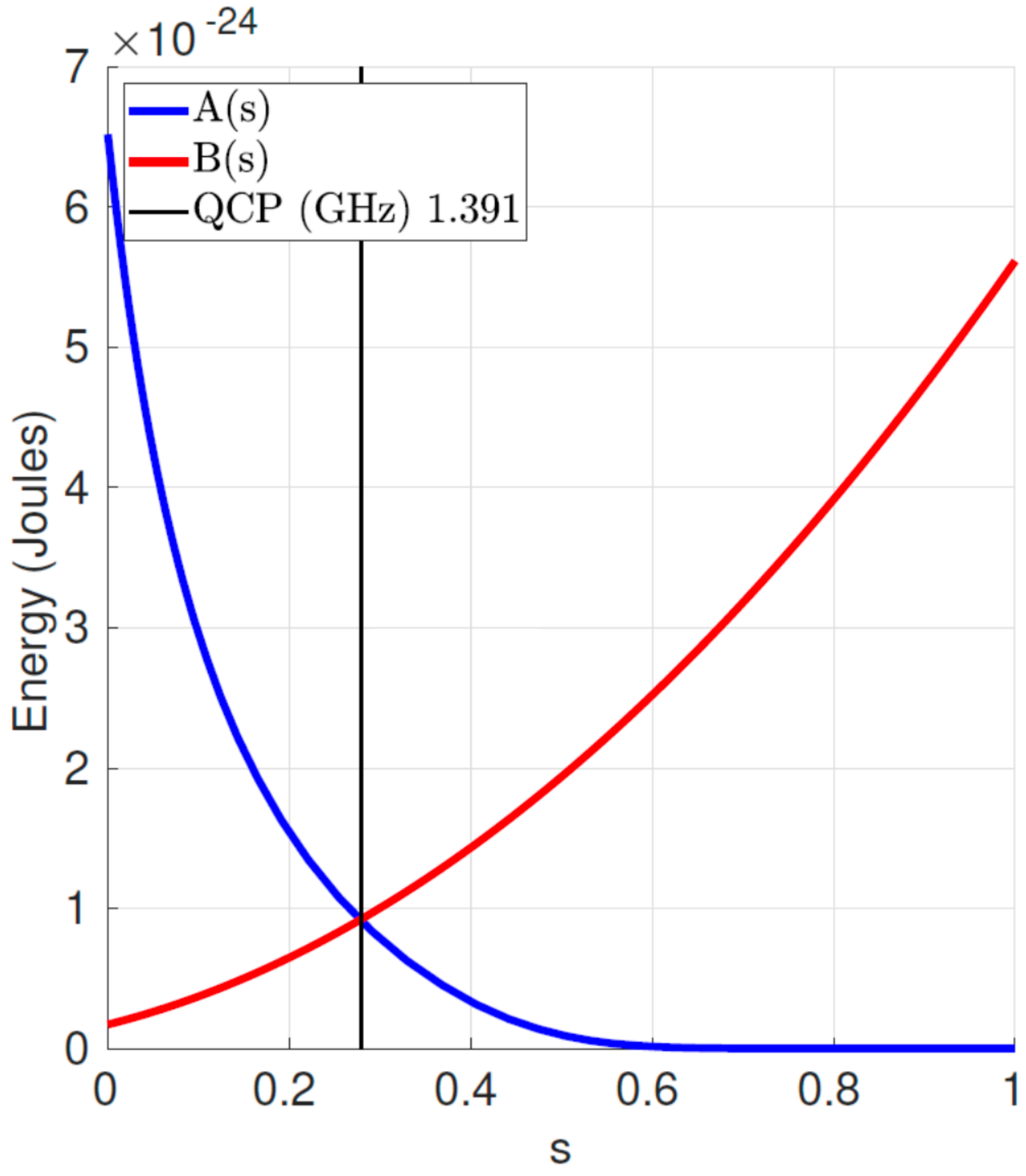
Conclusion

☞ Direct mapping of optimization tasks to current quantum hardware isn’t feasible. Generic embeddings exist, but worst-case problem instances constitute a major bottleneck.

☞ Threshold method: structure-aware pre-processing step that mitigates embedding’s effects

☞ Ultimately, the cut problems converge better to their target state, but there's a *trade-off* between **convergence** and **accuracy**.

Backup slide: D-Wave's annealing schedules



Backup slide: Minor embedding

☞ Represent a highly connected qubit using a linear chain of multiple qubits

☞ Overhead in number of variables

$$\mathcal{L}(\alpha) = \min_{\beta \in \{0,1\}^M} \mathcal{L}_2(\alpha, \beta)$$

☞ General framework, technicalities can be found in:

V. Choi, Minor-Embedding in Adiabatic Quantum Computation: I. The Parameter Setting Problem, Available: http://arxiv.org/abs/0804.4884

V. Choi, Minor-embedding in adiabatic quantum computation: II. Minor-universal graph design, Available: http://arxiv.org/abs/1001.3116